Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.13	20.10	
worker nodes in service, mean	6.00	6.00	-0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.01	-0.2%
power (W), mean	629	635	+0.9%
power (W), peak	652	672	+3.0%
energy (Wh)	211	213	+0.7%
energy, parked workers still on at idle power (Wh)	211	213	+0.7%
CPU used (cores), mean	0.898	1.05	+17.4%
CPU allocatable (cores), mean	24.00	24.00	-0.0%
utilisation (used / allocatable)	0.037	0.044	+17.4%
energy per core-hour (Wh)	701	602	-14.1%
node-hours per core-hour	6.68	5.69	-14.8%
pending pods, pod-minutes	0.000	0.000	0
pending pods, peak	0.000	0.000	0
HPA replicas, mean	9.11	9.07	-0.4%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.533	0.800	+50.0%
requests timed	1,035	1,130	+9.2%
time over the response line (% of samples)	1.45	0.000	-100.0%
response time (ms), mean	160	64.77	-59.5%
response time (ms), median	130	57.45	-55.7%
response time (ms), 95th percentile	338	94.81	-72.0%
response time (ms), 99th percentile	535	118	-77.9%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.2898	6.4720	-11.2%	[-2.5895, +0.9503]	EVEN, no significant difference
kwh_per_core_hour	0.7613	0.6799	-10.7%	[-0.2498, +0.0937]	EVEN, no significant difference
utilisation	0.0370	0.0431	+16.6%	[-0.0048, +0.0179]	EVEN, no significant difference
pending_pod_minutes_per_hour	0.0000	0.0000	n/a	[+0.0000, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	1.4877	2.2616	+52.0%	[-2.1780, +3.7313]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.392 0.42 0.445 0.467 0.486 0.502 0.517 0.531 0.542 0.552 0.558 0.563 0.568 0.573 0.577 0.581 0.585 0.586 0.588
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	621	630	6.00	4.50	0.000	0.000	0.667	0.918

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	617	619	6.00	6.00	0.000	0.000	0.576	0.601
2	6.00	6.00	619	620	6.00	6.00	0.000	0.000	0.585	0.639
3	6.00	6.00	620	622	6.00	6.00	0.000	0.000	0.649	0.790
4	6.00	6.00	635	639	9.00	9.50	0.000	0.000	1.07	1.15
5	6.00	6.00	634	642	9.67	10.00	0.000	0.000	1.03	1.23
6	6.00	6.00	633	644	10.00	10.00	0.000	0.000	0.993	1.29
7	6.00	6.00	646	662	10.00	10.00	0.000	0.000	1.34	1.76
8	6.00	6.00	649	664	10.00	10.00	0.000	0.000	1.41	1.82
9	6.00	6.00	647	658	10.00	10.00	0.000	0.000	1.36	1.66
10	6.00	6.00	636	646	10.00	10.00	0.000	0.000	1.08	1.37
11	6.00	6.00	619	616	10.00	10.00	0.000	0.000	0.641	0.560
12	6.00	6.00	617	619	10.00	10.00	0.000	0.000	0.553	0.626
13	6.00	6.00	624	627	10.00	10.00	0.000	0.000	0.763	0.817
14	6.00	6.00	634	641	10.00	10.00	0.000	0.000	1.03	1.21
15	6.00	6.00	637	640	10.00	10.00	0.000	0.000	1.08	1.15
16	6.00	6.00	633	643	10.00	10.00	0.000	0.000	0.998	1.25
17	6.00	6.00	628	629	10.00	10.00	0.000	0.000	0.849	0.896
18	6.00	6.00	620	620	10.00	10.00	0.000	0.000	0.647	0.651
19	6.00	6.00	617	616	10.00	10.00	0.000	0.000	0.611	0.563

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.